

ANSH MARU

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EDUCATION

University of Waterloo

Bachelor of Mathematics in Computational Mathematics - Honours

Waterloo, Ontario

Expected Graduation: 2029

- **Relevant Coursework:** Data Structures and Algorithmic Problem Solving, Object Oriented Software Development, Functional Programming, Computer Systems

EXPERIENCE

Patents & Innovation Intern

Pratt & Whitney Canada

Sept 2025 – Dec 2025

Mississauga, Ontario

- Performed open-domain research into **electronic engine control systems** supporting future projects
- Collaborated with engine controls team performing research into **embedded engine control software and hardware security**
- Deployed an internal PowerBI dashboard used by a cross-functional team of over **30+** members

PROJECTS

slangify | Python, Gemini API, Urban Dictionary API, Docker

- Built and deployed an always-on Discord bot for explaining slang, regional dialect, jargon, and non-standard language; supports **30+ users** and **70+ lookups** per day.
- Added selectable Gemini-only and Urban Dictionary-only modes, Discord embed formatting, and automated tests for parsing, formatting, and tokenization.
- Containerized and deployed to **Azure Cloud Services** using **Docker**.
- Reduced cloud hosting cost from roughly \$25/month to **\$5/month (80%)** by migrating and right-sizing the Azure deployment.

Neural Network | C++

- Engineered a modular multilayer perceptron framework by implementing custom layer and activation interfaces in **C++**, resulting in flexible support for arbitrary network depth.
- Assembled configurable activation modules (ReLU, Leaky ReLU or Sigmoid) selectable per layer to optimize gradient flow and convergence.
- Verified performance on the MNIST dataset, trained via mini-batch gradient descent with backpropagation, and achieving **95%** accuracy when tested.

ExSAMine | NextJS, FastAPI, Gemini API, Modal

- **Runner up for Best Digital Forensics Hack** at HoyaHacks 2026
- Developed AI-powered digital forensics platform using **Next.js** and **FastAPI**, integrating Google Gemini and **Meta's SAM3** for automated evidence analysis
- Built 3-stage AI pipeline: LLM-generated detection prompts to computer vision segmentation to forensic insights generation with confidence scoring
- Deployed object detection model to **Modal serverless GPUs** with automatic scaling and persistent model caching

Translatify | Electron, React, Typescript, Spotify API, DeepL API

- Built a desktop app that follows the current Spotify track and displays **time-synced lyrics with line-by-line translations**.
- Implemented **Spotify OAuth login with PKCE, token refresh**, and **playback polling** to keep the lyric display aligned with the song.
- Integrated LRCLIB lyric lookup and **DeepL translation**, including fallback search, batched requests, and cancellation when tracks change.
- Added persistent caching to avoid repeated lyric and translation API calls for previously viewed tracks.

TECHNICAL SKILLS

Languages: C++, C, C#, Typescript, Javascript, SQL, Python, HTML/CSS

Frameworks: Node.js, Flask, FastAPI, React, NextJS, Electron

Tools: Git, Github, Valgrind, gdb, Make, Vim, VS Code, Bash, Zsh, Linux, SQLite, PostgreSQL, $\LaTeX$